ACTIVITY ARRAYS AND STRINGS



PREVIOUS LECTURE

- Basics of Arrays
- Declaring Arrays
- Initializing Arrays
- Accessing Arrays
- Processing Arrays

Previous Lecture

Two Dimensional Arrays

String class in C++

C Style Strings

Outline

Activities for 1D Arrays

Activities for 2d Arrays

Activities for Strings

SINGLE DIMENSIONAL ARRAYS

Activity 1

a) What's wrong with the following array definitions?

- int readings[-1];
- float measurements[4.5];
- int size;
- string names[size];

b) What is "array bounds checking"? Does C++ perform it?

SOLUTION: ACTIVITY 1

a)

- Size can’t be -1
- Size must be constant + integer.
- Not array
  - Size is not

declare b)

- Accessing array elements using indexes beyond the boundaries (e.g., myList[-1] and myList[10]) causes an out-of-bounds error. Out of bounds is a serious error. Unfortunately, the C++ compiler does not report it. Be careful to ensure that array indexes are within bounds.

ACTIVITY 2

a)Which of the following statements are valid array declarations?

- double d[30];
- char[30] r;
- int i[] = (3, 4, 3, 2);
- float f[] = {2.3, 4.5, 6.6};

b)Suppose you have two arrays, list and myList. Can you copy myList to list using a syntax like the following?

list = myList;

SOLUTION: ACTIVITY 2

a)

- d[3];
- char[30] r;
- int i[] = (3, 4, 3, 2);
- float d[] = {2.3, 4.5, 6.6};

- b) This is not allowed in C++. You must copy individual elements from one array to the other as follows:

```
for (int i = 0; i < ARRAY_SIZE; i++)
{
list[i] = myList[i];
}
```

2D ARRAYS IN C++

Choose the best answer

a)

```
#include<iostream>
#include<string.h>
using namespace std;
main()
{
double[][] values =
   { {1.2, 9.0, 3.2},
     {9.2, 0.5, 1.5, -1.2},
     {7.3, 7.9, 4.8} } ;
}
```

What is the value of double [2][1] values?

1) 7.3
2) 7.9
3) There is no such array

2D ARRAYS IN C++

- Find and fix Error

```
#include<iostream>
#include<string.h>
using namespace std;
main()
{
double values[][] =
  { {1.2, 9.0, 3.2},
    {9.2, 0.5, 1.5, -1.2},
    {7.3, 7.9, 4.8} } ;



}
```

SOLUTION

- Necessary to specify the number of dimensions except the first.

2D ARRAYS IN C++

Write a C++ program that:

Takes the **number of rows and columns** from the user.

Takes input for a **2D array (matrix)**.

Displays the matrix in **row-wise format**.

Calculates and prints the **sum of all elements** in the matrix

```
for(i=0;i<rows;i++)
  for(j=0;j<columns;j++)
cout<<sum=sum+array[i][j]
```

SOLUTION

```
#include <iostream>
using namespace std;
int main()
{
   int rows, cols;

   // Step 1: Input rows and columns
   cout << "Enter number of rows: ";
   cin >> rows;

   cout << "Enter number of columns: ";
   cin >> cols;
```

```
// Step 2: Declare 2D array
   int matrix[10][10];   // assuming max size

   // Step 3: Input elements
   cout << "\nEnter elements of the matrix:\n";
   for (int i = 0; i < rows; i++)
   {
      for (int j = 0; j < cols; j++)
      {
         cin >> matrix[i][j];
      }
   }

   // Step 4: Display matrix (row-wise)
   cout << "\nMatrix (Row-wise):\n";
   for (int i = 0; i < rows; i++)
   {
      for (int j = 0; j < cols; j++)
      {
         cout << matrix[i][j] << " ";
      }
      cout << endl;
```

```
// Step 5: Calculate sum
   int sum = 0;
   for (int i = 0; i < rows; i++)
   {
      for (int j = 0; j < cols; j++)
      {
    cout<< “the sum is:”<<  sum = sum +
matrix[i][j];
      }
   }

   // Step 6: Display sum
   cout << "\nSum of all elements = " <<
sum << endl;

   return 0;
}
```